

OpenHCS: one editable workflow from images to results

A Choose how to work: UI editors and MCP bridge



Desktop forms



Edit



Python code



Edit



Agent conversation



Edit



CellProfiler import



Import

SHARED PIPELINE • configuration + ordered function steps



Named images

Prepare

Python function

Segment

Python function

Measure

Python function

Choose stack axes, processing groups,
function chains and named results

Submit workflow

B Connect data and processing tools



Image folders • Bio-Formats



OME-Zarr • OMERO (experimental)



Scientific Python + custom functions



C Separate execution and viewer processes

Worker processes

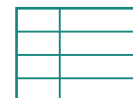
ZMQ server



CPU

GPU

Separate napari / Fiji viewers



Images • ROIs • tables

Catalog • compile • schedule

Requests ↔ status

Prepared tasks ↔ progress